Numerical Analysis

Lab 6

Giuseppe Caliò

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Iterative methods for linear systems

This lab deals with the numerical solution of a linear system

$$\mathbb{A}\boldsymbol{x} = \boldsymbol{b}, \quad \text{for } \mathbb{A} \in \mathbb{R}^{n\times n}, \boldsymbol{b} \in \mathbb{R}^n$$

We will focus on stationary iterative methods.

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Fill-in of a matrix I

Definition

"In general, the LU factorization process does not preserve the sparsity pattern of the original matrix A, as it may generate non-null elements in L and U in some positions (i, j) where the corresponding original elements aij were null. This phenomenon is called fill-in and depends on both the original pattern of A and the values of its entries" (A. Quarteroni et al., Scientific Computing with MATLAB and Octave)

Fill-in of a matrix II

Exercise 6.1 (Exam July 20, 2023)

Consider the following square matrix $\mathbb{A} \in \mathbb{R}^{n\times n}$:

$$\mathbb{A} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 1 \\ 0 & 1 & 0 & \dots & 0 & 1 \\ & & \ddots & & 0 & \vdots \\ 0 & & \dots & 1 & 0 & 1 \\ 0 & & \dots & 0 & 1 & 1 \\ 1 & 2 & \dots & 5 & 6 & 7 \end{bmatrix}.$$

Fill-in of a matrix III

Exam July 20, 2023

1. Using the Matlab commands `diag` and `ones`, construct the matrix $\mathbb{A}$. Provide the Matlab commands used.
2. Using the Matlab command `lu`, calculate the LU factorization of the previously constructed matrix. Provide the obtained factors $\mathbb{L}$ and $\mathbb{U}$, along with the Matlab commands used.
3. Using the Matlab command `spy`, answer the following questions, providing reasoning: Was pivoting performed? Did fill-in occur? Provide the Matlab commands used.
4. Given the known term $\boldsymbol{b} = [1, 0 \dots 0]^T \in \mathbb{R}^n$, solve the linear system $\mathbb{A}\boldsymbol{x} = \boldsymbol{b}$ using the functions `forwardsubstitution.m` and `backwardsubstitution.m`. Provide the obtained solution and the Matlab commands used.
5. Replace the last row of matrix $\mathbb{A}$ with the vector $\boldsymbol{v} = [1.1, 1 \dots 1] \in \mathbb{R}^n$ and repeat steps a, b and d for n=20,30,40. Knowing that the exact solution is $\boldsymbol{x} = [1 - \frac{1.1}{1.1+n-3}, -\frac{1.1}{1.1+n-3}, \dots - \frac{1.1}{1.1+n-3}, -\frac{1.1}{1.1+n-3}]^T \in \mathbb{R}^n$, calculate the relative error of the solution for each n.(Hint: use the Matlab command `ones` to create the components frome the second to the second-to-last element). Using the Matlab command `lu`, calculate the LU factorization of the new matrix. Provide the obtained error, along with the Matlab commands used.

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Richardson Method

Richardson methods consist in computing the sequence

$$\boldsymbol{x}^{(k)} = \boldsymbol{x}^{(k-1)} + \alpha\, \mathbb{P}^{-1}\, \boldsymbol{r}^{(k-1)}$$

where $\mathbb{P}$ is a suitable non singular **preconditioning matrix**, $\boldsymbol{r}^{(k-1)} = \boldsymbol{b} - \mathbb{A}\boldsymbol{x}^{(k-1)}$ is the **residual** vector associated with the $(k-1)$-th iteration and $\alpha > 0$ is the **acceleration parameter** of the method.

Richardson Methods

Richardson schemes are a generalization of the methods already presented,

$$\boldsymbol{x}^{(k+1)} = \mathbb{B}\boldsymbol{x}^{(k)} + \boldsymbol{g},$$

where the iteration matrix is

$$\mathbb{B}(\alpha) = \mathbb{I} - \alpha\mathbb{P}^{-1}\mathbb{A}$$

Convergence properties depend on the coefficient matrix $\mathbb{A}$ and method parameters α and $\mathbb{P}$.

- $\alpha = 1$, $\mathbb{P} = \mathbb{D}$: Jacobi ($\mathbb{B}_J = \mathbb{I} - (\mathbb{D})^{-1}\mathbb{A}$)
- $\alpha = 1$, with $\mathbb{P} = \mathbb{D} - \mathbb{E}$: Gauss-Seidel ($\mathbb{B}_{GS} = \mathbb{I} - (\mathbb{D} - \mathbb{E})^{-1}\mathbb{A}$)

Convergence of Richardson Methods

Theorem 6.1

For any non singular preconditioning matrix $\mathbb{P}$, the stationary Richardson method is convergent for any initial guess $\boldsymbol{x}^{(0)}$ iff

$$\frac{2Re(\lambda_i)}{\alpha|\lambda_i|^2} > 1, \quad \forall i = 1. \dots, n,$$

where $\lambda_i \in \mathbb{C}$ are the eigenvalues of $\mathbb{P}^{-1}\mathbb{A}$.

The result of this theorem comes from asking the spectral radius of the iteration matrix being less than one.

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Choice of α in the stationary Richardson method: optimality

Theorem 6.2

Let $\mathbb{A}$ and $\mathbb{P}$ be symmetric positive definite matrices. Then stationary Richardson methods are convergent for any initial guess if and only if the parameter α is selected in the $[0, 2/\lambda_{\max}]$ interval, being $\lambda_{\max}$ the maximum eigenvalue of $\mathbb{P}^{-1}\mathbb{A}$. Moreover the spectral radius of the iteration matrix is minimized for $\alpha = \alpha_{opt}$, where

$$\alpha_{opt} = \frac{2}{\lambda_{\min} + \lambda_{\max}}.$$

and $\lambda_{\min}$ the minimum eigenvalue of $\mathbb{P}^{-1}\mathbb{A}$. The spectral radius of the iteration matrix $\mathbb{B}(\alpha)$ is minimum if $\alpha = \alpha_{opt}$, with

$$\rho_{opt} = \frac{\lambda_{max} - \lambda_{min}}{\lambda_{min} + \lambda_{max}}.$$

Choice of α in the stationary Richardson method: optimality

Exercise 6.2

Consider the linear system $\mathbb{A}\boldsymbol{x} = \boldsymbol{b}$, with $\boldsymbol{b} = 0.2{*}$**ones(50,1)** and

$$\mathbb{A} = \begin{bmatrix} 4 & -1 & -1 & & & \\ -1 & 4 & -1 & -1 & & \\ -1 & -1 & 4 & -1 & -1 & \\ & \dots & \dots & \dots & \dots & \\ & & -1 & -1 & 4 & -1 \\ & & & -1 & -1 & 4 \end{bmatrix}, \quad \mathbb{T} = \begin{bmatrix} 2 & -1 & & & & \\ -1 & 2 & -1 & & & \\ & -1 & 2 & -1 & & \\ & \dots & \dots & \dots & \dots & \\ & & & -1 & 2 & -1 \\ & & & & -1 & 2 \end{bmatrix}.$$

a. Are $\mathbb{A}$ and $\mathbb{T}$ symmetric positive definite matrices?
b. Apply the Richardson method starting from $\boldsymbol{x}^{(0)} = \mathbf{0}$, with a tolerance equal to 10^{-6} and $\mathbb{P} = \mathbb{I}$. Compare the required number of iterations (if the method is convergent) for $\alpha = 0.2$, $\alpha = 0.33$ and $\alpha = \alpha_{opt}$.
c. Apply the Richardson method starting from $\boldsymbol{x}^{(0)} = \mathbf{0}$, with a tolerance equal to 10^{-6}, $\mathbb{P} = \mathbb{T}$, $\alpha = \alpha_{opt}$. Compare the required number of iterations with the one associated with the non-preconditioned case. Compare the condition numbers of $\mathbb{A}$ and $\mathbb{P}^{-1}\mathbb{A}$.

Choice of α in the stationary Richardson method: optimality

Exercise 6.3

Consider

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 & 1 & \dots & 1 \\ 4 & 4 & 1 & 1 & \dots & 1 \\ 8 & 8 & 8 & 1 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & 1 \\ 2^n & \dots & \dots & \dots & \dots & 2^n \end{bmatrix} \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ \vdots \\ n \end{bmatrix} \qquad \text{con } n = 20.$$

Solve the linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ by using three different preconditioned stationary Richardson schemes: $\mathbf{P}_1$: (no preconditioner), $\mathbf{P}_2$: (Jacobi), $\mathbf{P}_3$: (Gauss-Seidel). Set `toll=1e-6`, `nmax=1e3`, $\mathbf{x}_0 = [1, 1, \dots, 1]^T$. Select the acceleration parameter α as the midpoint of the convergence interval. *Hint*: in case of **complex eigenvalues** of the preconditioned matrix, if their real part is always positive, then the method converges for $0 < \alpha < 2 \min \frac{Re(\lambda_i)}{|\lambda_i|^2}$.

Exam problem I

Exercise 6.4 (Exam July 11, 2024)

Consider the linear system $\mathbb{A}\boldsymbol{x} = \boldsymbol{b}$, with $\mathbb{A} \in \mathbb{R}^{20\times 20}$, symmetric positive definite, and $\boldsymbol{b} \in \mathbb{R}^{20}$ given by $\mathbb{A} =$`pentadiag(-1,-4,10,-4,-1)` and $\boldsymbol{b} = [1, 1 \ldots 1]^T$. Here, `pentadiag` refers to a pentadiagonal matrix, meaning it has only the 5 central diagonals different from zero, filled with the given values.

Exam problem II

Exam July 11, 2024

1. Solve the proposed system using the stationary Richardson method with $\mathbb{P} = \mathbb{I}$(use the function `richardson.m`), $\alpha = 0.25$ initial guess $\boldsymbol{x}^{(0)} = \boldsymbol{b}$, tolerance $tol = 10^{-3}$ and $n_{max} = 500$. Report the number of iterations N performed, the third component $\boldsymbol{x}_3^{(N)}$ of the approximated solution vector $\boldsymbol{x}^{(N)}$ and the value of the corresponding relative residual $\boldsymbol{r}_3^{(N)}$. Use format long. Also, provide the Matlab instructions used.
2. Comment on the results obtained in the previous point, providing appropriate theoretical explanations on how such considerations could have been anticipated a priori.
3. Rerun the method as in point a, but now with the optimal value of the parameter α_{opt}. Report this value, the number of iterations required by the method, a comment on the results, and the Matlab instructions used.
4. Now solve the problem from point a using $\alpha = 0.84$ with the preconditioned Richardson method with $\mathbb{P} = tridiag(-5, 10, -5)$ Here, tridiag refers to a tridiagonal matrix, meaning it has only the 3 central diagonals different from zero, filled with the given values. Provide the Matlab instructions, the number of iterations used, and a comment on the results.

Exercise 6.5

Consider the following matlab commands:

```
n=20; A=eye(n);
A(1,:)=A(1,:)-1/n/10;
A(:,1)=A(:,1)-1/n/10;
b=b(2:n,1) = (n*10 + 1)/(n*10);
b(1) = (11*n + 1)/(n*10);
xex = ones(n,1);
```

1. compare the sparsity pattern of the exact (LU) and inexact (ILU) factorization;
2. Check the convergence of the stationary Richardson method, adopting P=I and P equal to the incomplete factorization of A and then solve the system; Set $\alpha = 1.9$, $x^{(0)} = 0$, $tol = 1e-3$, $nmax = 200$. Compare the number of iterations and compute the relative error of the solution;
3. Repeat the previous point with the optimal value of alpha;

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Homework I

Homework 6.1

Consider the following linear system $\mathbb{A}_n \boldsymbol{x} = \boldsymbol{b}_n$ with $n = 10, 15$, where matrix $\mathbb{A}_n$ and vector $\boldsymbol{b}_n$ are constructed with the MATLAB statements

$$\mathbb{A}_n = \texttt{eye(n) + triu(pascal(n),1)}, \quad \boldsymbol{b}_n = \texttt{[1:n]'}.$$

Consider now the following family of iterative methods with parameter α

$$\boldsymbol{x}^{(k+1)} = (\mathbb{I} + \alpha \mathbb{A}_n)\, \boldsymbol{x}^{(k)} - \alpha \boldsymbol{b}_n$$

where $\mathbb{I}$ is the identity matrix of order n.

- a. Use `MATLAB` to determine if the methods obtained by setting $\alpha = 1, -1/2$ and -1 are convergent to the exact solution $\boldsymbol{x}^*$ of the system, in both cases $n = 10, 15$.
- b. Implement the method for the cases in which convergence is assured. Find the solution $\boldsymbol{x}^*$ with tolerances 10^{-6} and 10^{-9} for both $n = 10, 15$ and indicate the number of performed iterations. Report the solutions in `MATLAB short e` format.
- c. What about the case of $\alpha = -1$? Can you motivate the result of the previous point?
- d. Was it possible to know the behaviour of the methods just looking at the structure of the iteration matrices $(\mathbb{I} + \alpha \mathbb{A}_n)$? How?

References I